

1. Regional price pressure and Botswana

The supplied Human Capital Project file contains FAO consumer-price indicators, not direct health, education or nutrition outcomes. Results therefore describe regional price-pressure linkages, not causal human-capital effects.

Method

We tested two distinct indicators from the supplied comparison file: regional food inflation and regional general CPI inflation. Each indicator is the South Africa-Namibia mean. We estimated a separate regression of Botswana food inflation on its own one-month lag and the indicator's one-month lag, then ran a two-lag incremental predictive test.

Two supplied indicator families

Indicator	Lag-1 beta	Reg. p	Granger F	Granger p
Regional food inflation	0.044	0.0291	6.446	0.0018
Regional general CPI inflation	0.032	0.2429	3.818	0.0232

Regional food inflation is the direct affordability channel: its lag coefficient is statistically significant. Regional general inflation represents wider cost-of-living pressure: its single lag is not significant, but its two lags are jointly predictive. These are conditional associations, not causal effects.

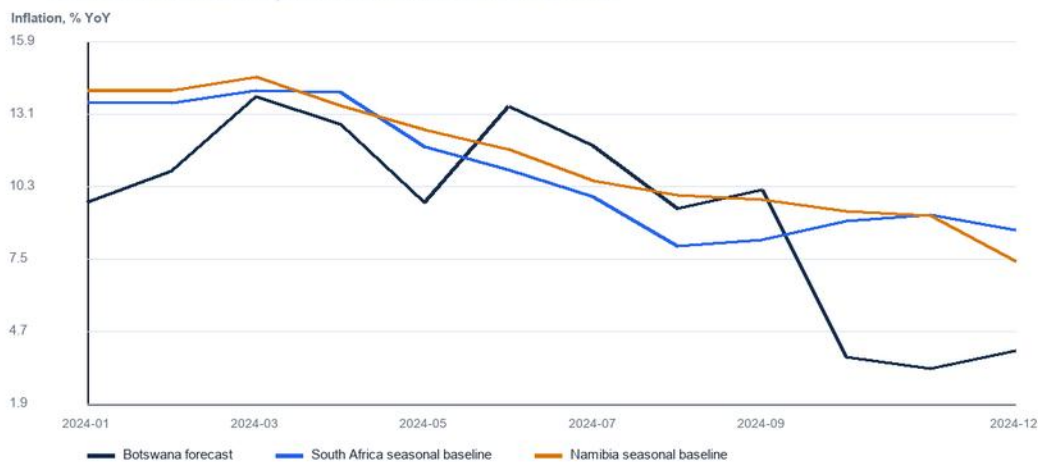
Country-level predictive checks

Driver	F statistic	p-value	Two-lag evidence
South Africa	8.027	0.0004	Yes
Namibia	3.063	0.0484	Yes

2. Interpretation and forward view

2024 regional price-pressure projection

Botswana model forecast; comparison countries use 2023 seasonal baselines



Quantified 2024 household-pressure projection

Our submitted forecast averages 9.4% year-on-year in 2024 and peaks at 13.8% in 2024-03. Food inflation is at least 10% in 6 of 12 months. The average eases from 11.7% in the first half to 7.1% in the second half, a 4.6-percentage-point reduction.

Human-capital interpretation and boundaries

- 6 double-digit-inflation months imply sustained food-affordability and diet-quality pressure, especially for lower-income households.
- The second-half easing reduces that pressure, but December inflation is still forecast at 3.9% year-on-year.
- The supplied data do not measure schooling, health or nutrition outcomes, so we report an exposure proxy rather than inventing a causal HCP effect.

Forward projection

Botswana's line is our Phase 1 forecast. The South Africa and Namibia comparison lines reuse their 2023 seasonal values and average 11.1% in 2024. No actual 2024 values enter this projection. As new observations arrive, policy monitoring should replace the seasonal baselines and track whether the 6 projected high-pressure months materialise.